

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

The de Broglie wavelength of a particle is given by the equation :

$$\hat{\lambda} = h / p$$

where $\hat{\lambda}$ is the de Broglie wavelength , h is the Planck's constant (6.626×10^{-34} J s), and p is the momentum of the particle .

The kinetic energy of an electron is given by :

$$KE = (1/2)mv^2$$

where KE is the kinetic energy , m is the mass of the electron (9.11×10^{-31} kg), and v is the velocity of the electron .

We can find the velocity of the electron using the given kinetic energy :

$$\begin{aligned} KE &= (1/2)mv^2 \\ 1\text{ eV} &= 1.6 \times 10^{-19}\text{ J} \\ (1.6 \times 10^{-19}\text{ J}) &= (1/2)(9.11 \times 10^{-31}\text{ kg})v^2 \\ v^2 &= 2(1.6 \times 10^{-19}\text{ J}) / (9.11 \times 10^{-31}\text{ kg}) \\ v^2 &= 3.51 \times 10^{11}\text{ m}^2/\text{s}^2 \\ v &= \sqrt{(3.51 \times 10^{11}\text{ m}^2/\text{s}^2)} = 5.92 \times 10^5\text{ m/s} \end{aligned}$$

Now we can find the momentum of the electron :

$$\begin{aligned} p &= mv \\ p &= (9.11 \times 10^{-31}\text{ kg})(5.92 \times 10^5\text{ m/s}) \\ p &= 5.39 \times 10^{-26}\text{ kg m/s} \end{aligned}$$

Now we can find the de Broglie wavelength :

$$\begin{aligned} \hat{\lambda} &= h / p \\ \hat{\lambda} &= (6.626 \times 10^{-34}\text{ J s}) / (5.39 \times 10^{-26}\text{ kg m/s}) \\ \hat{\lambda} &= 1.23 \times 10^{-7}\text{ m} \\ \hat{\lambda} &= 1.23 \times 10^{-7}\text{ cm} \end{aligned}$$

Therefore , the de Broglie wavelength is 1.23×10^{-7} cm

</think>< answer> G . 1.23×10^{-7} cm </answer>